Online Shoppers Purchasing Intention

Identifying the Drivers of Purchase (Revenue) from Browsing Behaviour

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1 Introduction

Online retailers can track almost everything a visitor does on a website, but only a small fraction of visits end in a purchase. This report looks at 12,205 browsing sessions from an online retailer, each described by 17 behavioural and contextual predictors, to understand which session characteristics are associated with a completed purchase. The outcome of interest, `Revenue`, is a binary flag that is `TRUE` for the 15.6% of sessions that resulted in a sale.

Because the outcome is binary, purchase is modelled with **logistic regression** rather than ordinary linear regression. The report proceeds in three stages: an exploratory look at how behaviour differs between purchasers and non-purchasers, the fitting and refinement of a logistic regression model (including an important data-leakage issue that had to be corrected), and an evaluation of how well the final model identifies likely purchasers on data it has not seen before.

2 Data Overview

The raw dataset contained 12,330 sessions, of which 125 were exact duplicates (the same session logged twice) and were removed, leaving 12,205 sessions for analysis. There is no missing data in this file.

Identifier-style columns (`OperatingSystems`, `Browser`, `Region`, `TrafficType`, `Month`, `VisitorType`) were treated as categorical, since their numeric codes carry no meaningful order or scale. All count, duration, and rate columns (e.g. `ProductRelated`, `BounceRates`, `PageValues`) were kept numeric.

One data-quality check is worth flagging: 592 sessions (4.9%) show at least one product page viewed but a total browsing duration of zero seconds – a timing inconsistency that is best explained by how the tracking script rounds very short visits, rather than by an invalid session. These rows were kept rather than dropped, since removing them would bias the sample toward slower browsers without a clear justification.

3 Exploratory Analysis

3.1 Purchase Behaviour by Session Characteristics

Table 1: Average session behaviour, by purchase outcome

Metric	No purchase	Purchase
Sessions	10297.00	1908.00
Admin. pages	2.14	3.39
Admin. time (s)	74.64	119.48
Info. pages	0.46	0.79
Info. time (s)	30.60	57.61
Product pages	29.05	48.21
Product time (s)	1082.98	1876.21
Bounce rate	0.02	0.01
Exit rate	0.05	0.02
Page value	2.00	27.26
Special day	0.07	0.02

Sessions that end in a purchase look, on average, like more engaged sessions: more product pages viewed, more time spent on the site, lower bounce and exit rates, and a much higher page value. None of this is surprising on its own – it simply confirms the dataset behaves the way an online-retail dataset should before any modelling is attempted.

3.2 Conversion Rate by Visitor Segment

Table 2: Conversion rate by visitor type

Visitor type	Sessions	Conversion rate (%)
New_Visitor	1693	24.9
Other	81	19.8
Returning_Visitor	10431	14.1

Table 3: Conversion rate: weekday vs. weekend

Weekend	Sessions	Conversion rate (%)
FALSE	9346	15.1
TRUE	2859	17.5

New visitors convert at roughly 24.9%, noticeably higher than returning visitors (14.1%), and weekend sessions convert slightly more often than weekday sessions, though that gap is modest. Conversion rate by month is easier to read as a chart than a table, since the goal is to compare twelve values against each other:

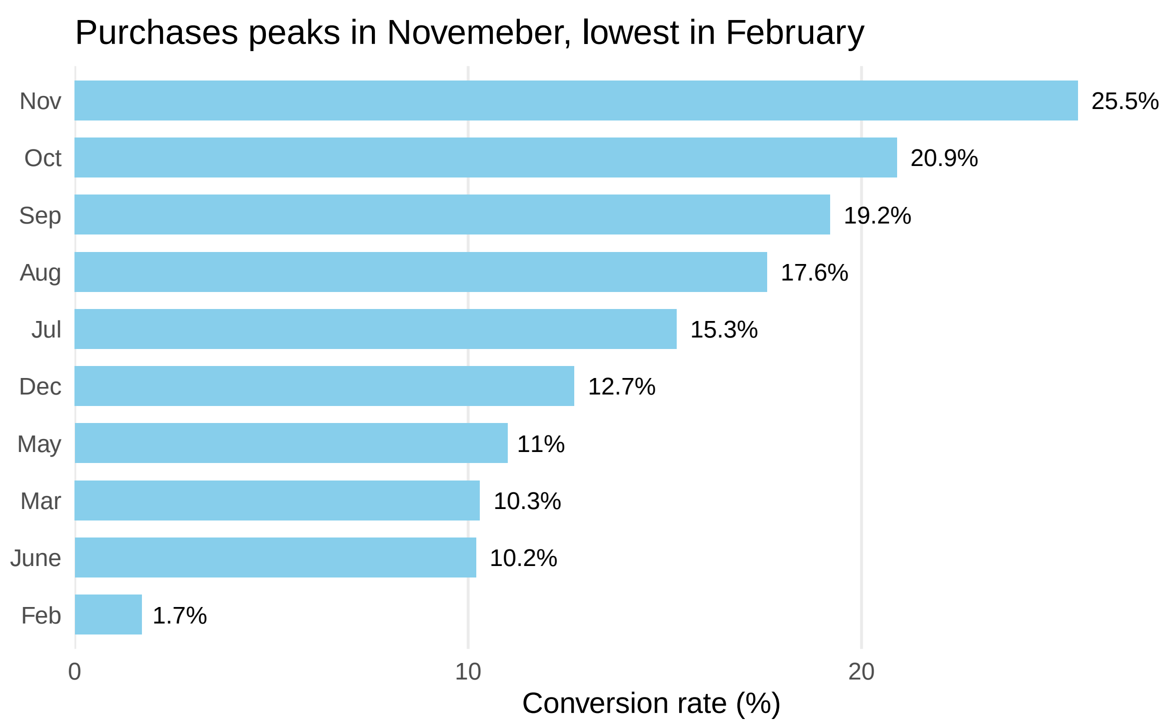
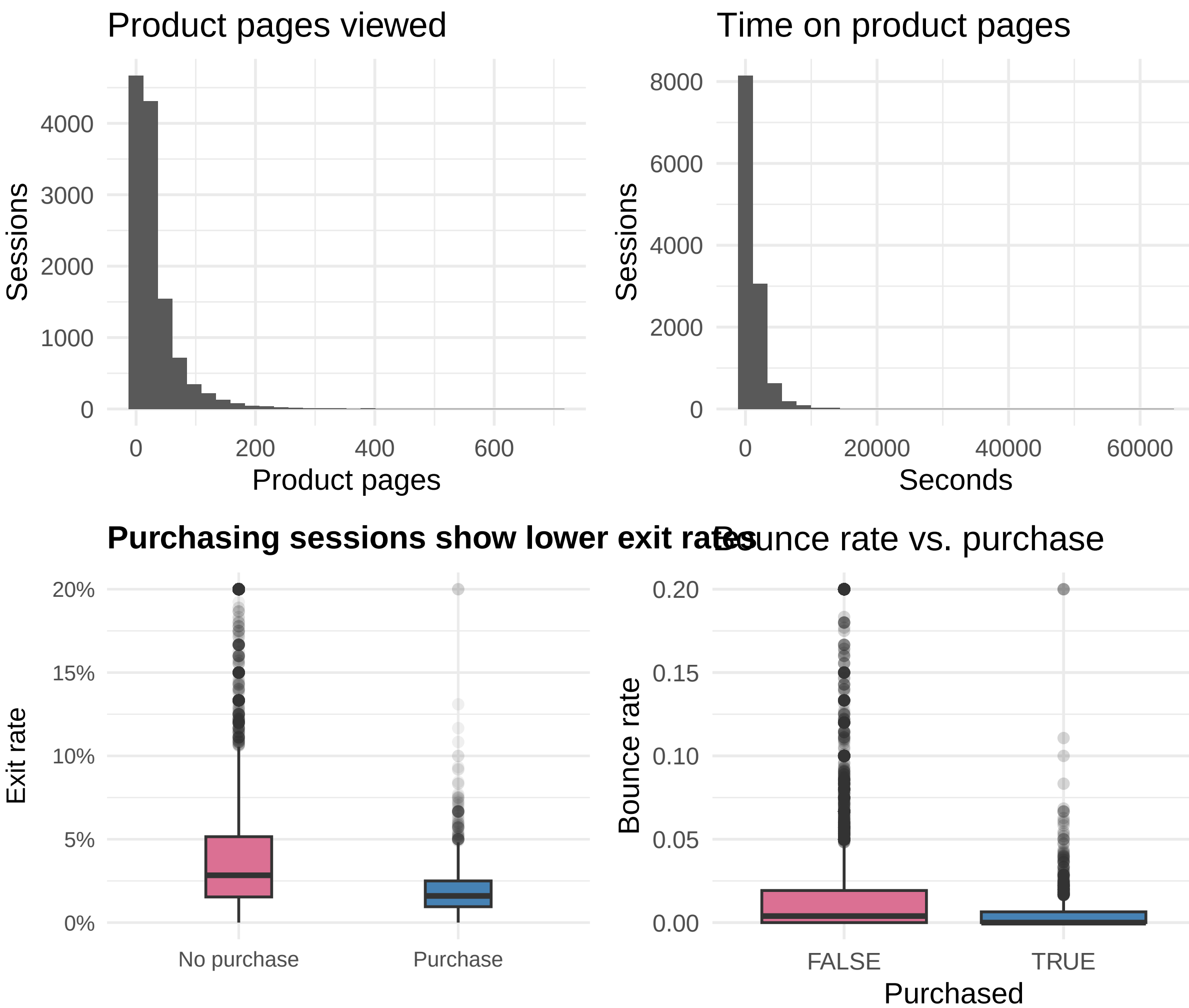


Figure 1: Conversion rate by month, highest to lowest

Nov stands out clearly as the strongest month for conversion (25.5%), consistent with a seasonal shopping effect, while February is the weakest month by a wide margin. These are descriptive patterns rather than causal claims – month and traffic source are correlated with many other factors (promotions, seasonal stock, ad spend) that this dataset does not capture.

3.3 Distribution of Key Behavioural Metrics



`ProductRelated` is heavily right-skewed: most sessions view only a handful of product pages, with a long tail of "power browsers" viewing hundreds. Exit rate shows a clean separation between purchasers and non-purchasers – purchasing sessions cluster near zero, while non-purchasing sessions spread much higher. Bounce rate tells a weaker story: both groups sit close to zero and overlap heavily, which foreshadows the multicollinearity check below.

3.4 Multicollinearity Diagnostics

Several of the numeric predictors describe closely related aspects of the same browsing behaviour (e.g. pages viewed and time spent viewing them), so a correlation check is needed before fitting a regression model – highly correlated predictors make it difficult to isolate which one is actually driving the outcome.

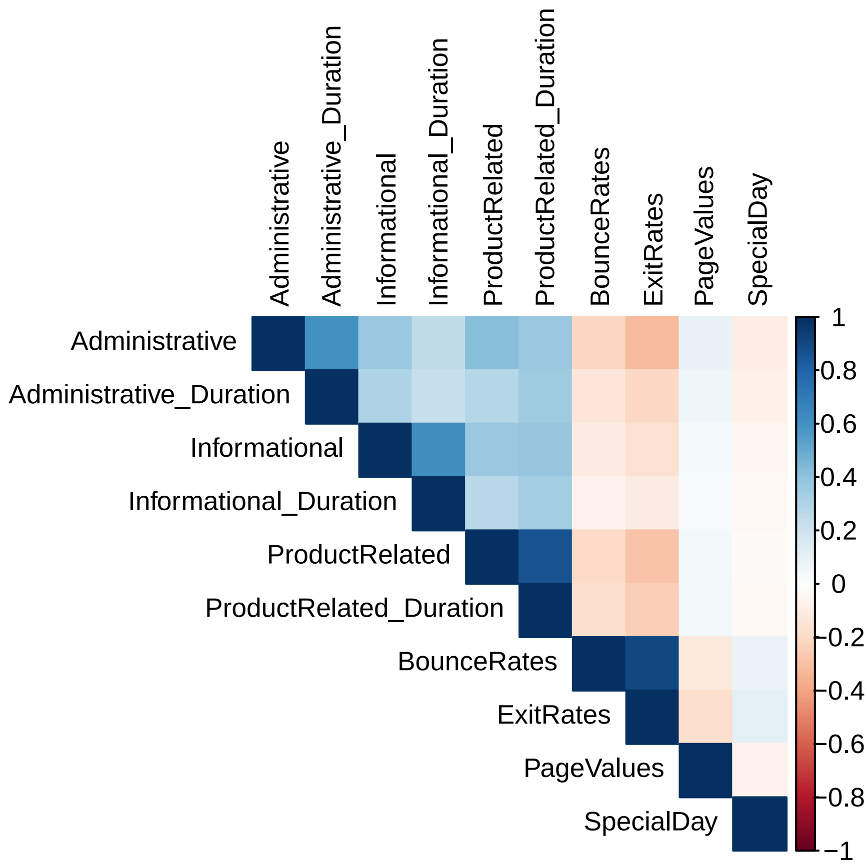


Figure 2: Correlation heatmap of numeric predictors

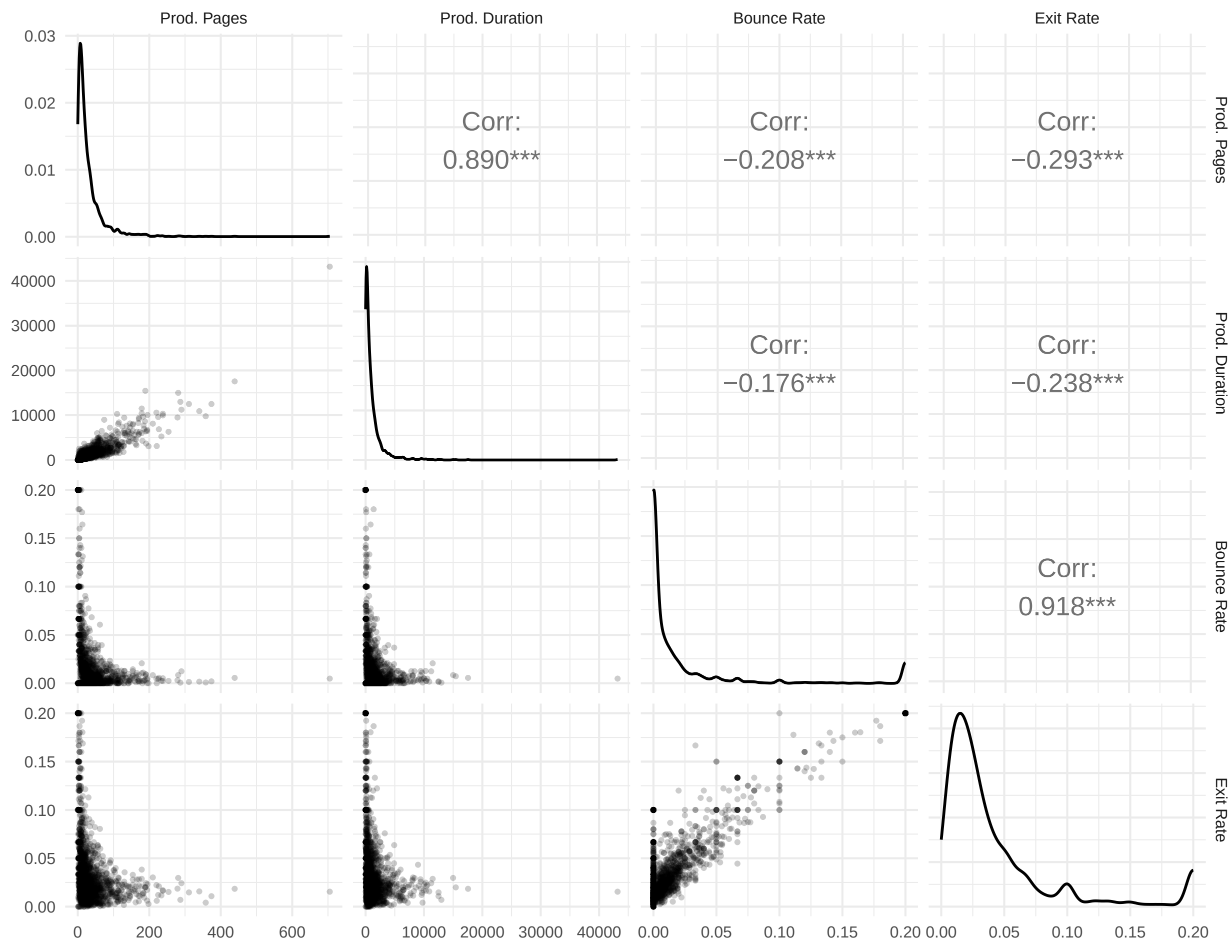


Figure 3: Pairwise relationships between the four most collinear predictors

Two pairs stand out as strongly collinear: `BounceRates` and `ExitRates` ($r = 0.9$), and `ProductRelated` and `ProductRelated_Duration` ($r = 0.86$). Including both members of a highly correlated pair in the same regression makes their individual coefficients unstable and hard to interpret, so one variable from each pair needs to be dropped before modelling.

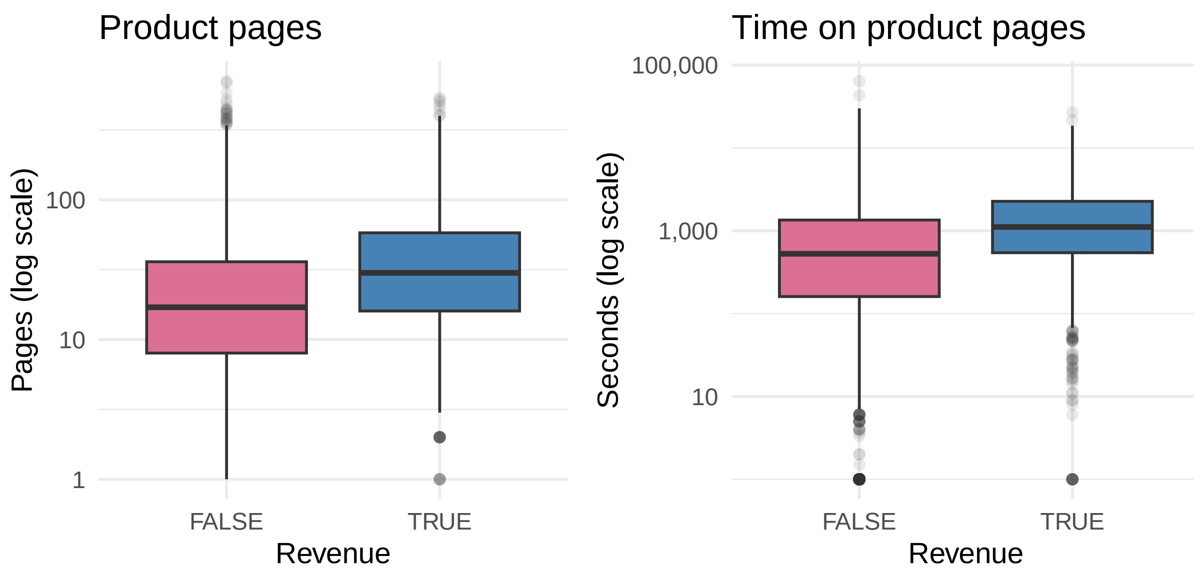


Figure 4: Product pages and time on product pages, by purchase outcome (log scale)

`ProductRelated` and `ProductRelated_Duration` separate purchasers from non-purchasers about equally well, so the simpler count variable, `ProductRelated`, is kept and `ProductRelated_Duration` is dropped. For the second pair, the exit-rate and bounce-rate boxplots shown earlier (Figure 2) already make the deciding case: `ExitRates` shows a much cleaner split between purchasers and non-purchasers, while `BounceRates`' median sits near zero for both groups and overlaps heavily – so `ExitRates` is kept and `BounceRates` is dropped.

4 Model Development

4.1 Train/Test Split

The data is split 70/30 into training and test sets. The model is fit only on the training set; the test set is held out entirely until final evaluation, so that reported performance reflects how the model behaves on sessions it has not seen.

`ExitRates` is stored as a proportion (0 to 0.2). It is rescaled to percentage points (0 to 20) before modelling, so that a one-unit change in the model corresponds to a one-percentage-point change in exit rate. This is purely a change of units: it does not affect model fit, significance, or predictions, but it makes the resulting coefficient directly interpretable.

```
set.seed(4187)
i <- sample(seq_len(nrow(df)), 0.7 * nrow(df))
train <- df[i, ]
test  <- df[-i, ]
```

Rare categories within `Browser`, `OperatingSystems`, and `TrafficType` (fewer than 100 sessions) are grouped into an "Others" level. This grouping is learned from the **training set only** and then applied to the test set, so that no information about the test set's category frequencies leaks into model training.

4.2 Initial Model and the PageValues Leakage Issue

A first logistic regression, using every remaining predictor, finds `ProductRelated`, `ExitRates`, and `PageValues` as significant numeric predictors, along with several categorical levels (particular months, traffic types, and visitor types). `PageValues` stands out immediately: it is by far the most significant term in the model, yet it barely correlates with any other predictor in the dataset – unusual, since almost every other behavioural variable is related to at least one other (more pages tends to mean more time on site and a lower exit rate).

Table 4: Conversion rate by whether PageValues is non-zero

Has page value	Sessions	Conversion rate
FALSE	9475	0.039
TRUE	2730	0.563

The reason becomes clear from Table 4: sessions with a non-zero page value convert at a vastly higher rate than sessions with a page value of zero. `PageValues` is a Google Analytics metric that is itself calculated from the pages a visitor viewed *before* completing a transaction – meaning it partially encodes the outcome being predicted. This is a textbook case of **data leakage**: including it would make the model look highly accurate without revealing anything actionable about visitor behaviour. `PageValues` is therefore excluded from the model used for interpretation and evaluation going forward.

4.3 Chi-Squared Test

Chi squared test is used to determine if there is statistically significant association between Month and Revenue.

```
##
##  Pearson's Chi-squared test
##
## data:  train$Month and train$Revenue
## X-squared = 282.74, df = 9, p-value < 2.2e-16
```

```
##            train$Revenue
## train$Month     FALSE      TRUE
##        Feb   104.1677  18.83226
##        Mar  1088.2565 196.74353
##        May  1977.4933 357.50673
##        June  172.7660  31.23399
##        Jul   248.1394  44.86059
##        Aug   257.4552  46.54477
##        Sep   270.1586  48.84139
##        Oct   321.8190  58.18097
##        Nov  1812.3493 327.65071
##        Dec   982.3949 177.60506
```

```
##            train$Revenue
## train$Month  FALSE   TRUE
##        Feb    4.25  -4.25
##        Mar    5.11  -5.11
##        May    7.05  -7.05
```

```
##        June   2.41  -2.41
##        Jul    0.64  -0.64
##        Aug   -0.72   0.72
##        Sep   -0.66   0.66
##        Oct   -1.87   1.87
##        Nov  -15.56  15.56
##        Dec    4.18  -4.18
```

Standardised residuals beyond +-2 fall outside the range expected under independence at roughly a 95% confidence level. Several months, most notably November at +15.56, exceed this threshold, confirming that Month and Revenue are not independent (282.7, $p < 0.001$). However, Cramer's V for this table is only 0.18, which indicates a small-to-moderate association. In other words, Month is a meaningful factor in purchase behaviour, but a considerably weaker one than the size of the individual residuals might suggest.

4.4 Final Model

Table 5: Model fit comparison (lower AIC is better)

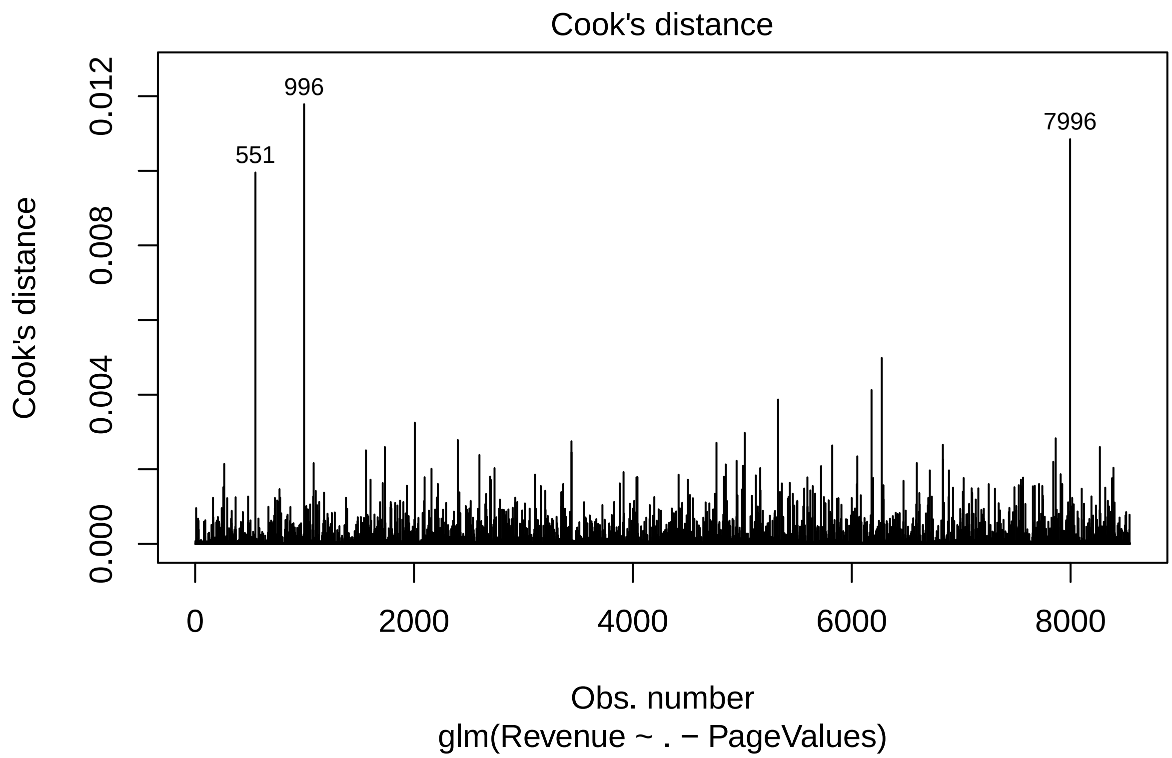
	df	AIC
model	51	5022.516
model_reduced	50	6380.286

As expected, removing `PageValues` costs a substantial amount of model fit (AIC rises by roughly 1358 points) – confirming how much explanatory power that single variable was contributing, disproportionate to its near-zero relationship with everything else. The remaining model, `model_reduced`, is the one used for interpretation and evaluation, since its results reflect genuine, actionable browsing behaviour.

Table 6: Variance inflation factors, final model

Predictor	GVIF	Df	GVIF^(1/(2*Df))
Administrative	1.80	1	1.34
Administrative_Duration	1.56	1	1.25
Informational	1.87	1	1.37
Informational_Duration	1.67	1	1.29
ProductRelated	1.45	1	1.21
ExitRates	1.17	1	1.08
SpecialDay	1.21	1	1.10
Month	1.98	9	1.04
OperatingSystems	10.13	4	1.34
Browser	8.39	7	1.16
Region	1.27	8	1.01
TrafficType	3.05	11	1.05
VisitorType	2.83	2	1.30
Weekend	1.04	1	1.02

Every adjusted GVIF value is comfortably below the range that would signal concerning multicollinearity, confirming that dropping `BounceRates` and `ProductRelated_Duration` earlier resolved the issue identified in the exploratory analysis.



The three largest Cook's distance values are all below 0.012 – well under the conventional threshold of 1 used to flag a genuinely influential observation. These sessions are not extreme enough to warrant removal; no rows are excluded on this basis.

4.5 Odds Ratios and Interpretation

Coefficients from a logistic regression are on the log-odds scale and are not directly comparable across predictors that sit on very different scales – `ExitRates` spans 0 to 20 percentage points, while `ProductRelated` is a count running into the hundreds. To interpret effect sizes fairly, coefficients are converted to odds ratios and expressed in meaningful units: per one percentage point for exit rate, and per ten additional pages for product-page views.

Table 7: Odds ratios for key numeric predictors (per one-unit increase)

Term	Odds ratio	CI (low)	CI (high)	p-value
Administrative	1.012	0.990	1.035	0.286
Informational	1.053	0.997	1.111	0.062
ProductRelated	1.002	1.001	1.004	0.002
ExitRates	0.747	0.716	0.779	<0.001
SpecialDay	0.556	0.328	0.910	0.024

On more interpretable units: each additional 10 product pages viewed is associated with about a 2.2% increase in the odds of purchase, holding everything else fixed. Each 1-percentage-point increase in exit rate is associated with roughly a 25% *reduction* in the odds of purchase – making exit rate the strongest and most consistent behavioural predictor in the final model. `SpecialDay` carries a smaller but statistically significant negative association. Several `TrafficType` levels and the months July through November are also significant relative to their reference categories, but with wider confidence intervals, reflecting smaller subgroup sizes – these contextual effects are noted, but should be treated as less reliable than the behavioural predictors above.

5 Model Evaluation

5.1 Classification Performance

Table 8: Test-set performance at the default 0.5 threshold

Metric	Value
Accuracy	0.835
Sensitivity (recall)	0.033
Specificity	0.992
Precision	0.465

At the conventional 0.5 cutoff, the model correctly classifies most sessions overall, but this is misleading: only 15.6% of sessions convert, so a model that predicted "no purchase" for everyone would already be right most of the time. The metric that matters here is sensitivity – of the sessions that actually converted, how many did the model catch? At 0.5, the answer is only 3.3%, meaning the model misses the large majority of real purchasers. This does not mean the model has no discriminating power; it means 0.5 is a poor cutoff for a dataset this imbalanced.

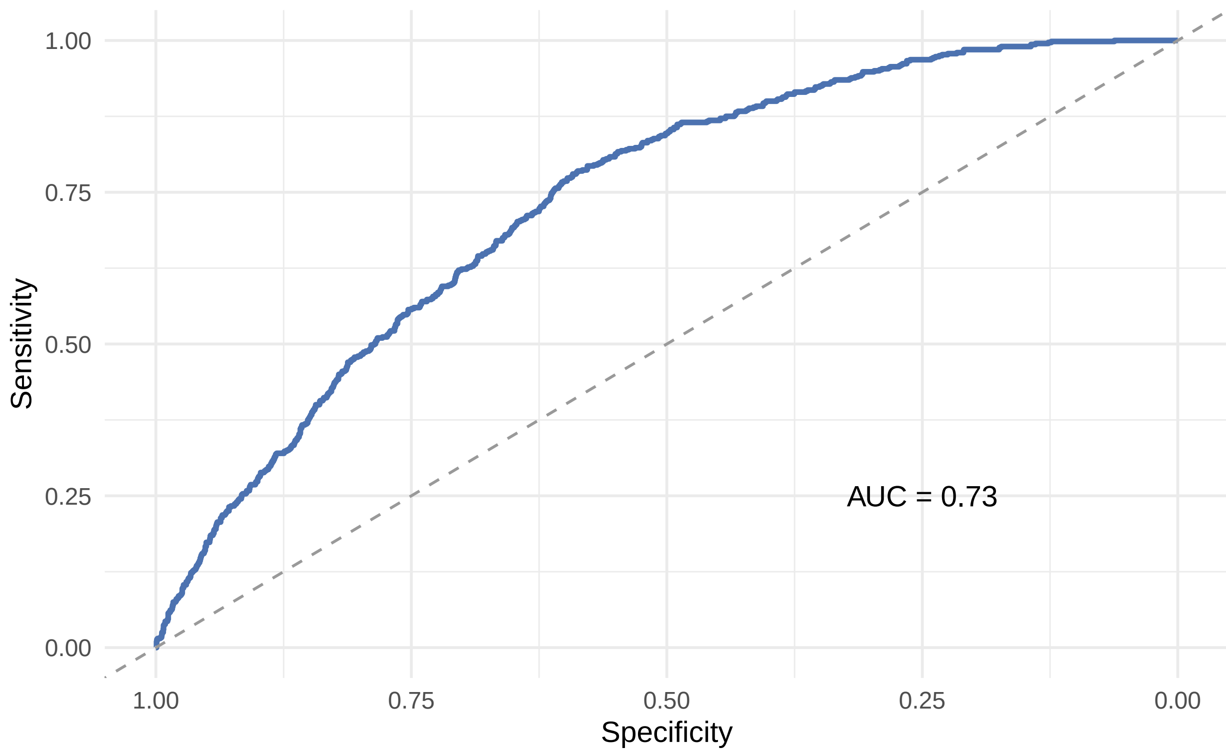


Figure 5: ROC curve, final model on held-out test data

The area under the ROC curve is 0.73: if one purchasing session and one non-purchasing session are chosen at random, the model assigns a higher predicted probability to the purchasing session about 73% of the time. This confirms the model has genuine, moderate discriminating power – independent of whatever cutoff is used to turn that probability into a yes/no prediction.

5.2 Choosing a Decision Threshold

Table 9: Threshold comparison: default cutoff vs. Youden-optimal cutoff

Threshold	Sensitivity	Specificity	Precision
0.50 (default)	0.033	0.992	0.465
0.13	0.785	0.587	0.271

Lowering the cutoff to 0.13 (the Youden-optimal point, which jointly maximises sensitivity and specificity) lifts sensitivity to 78%, at the cost of precision falling to 27% – more sessions get flagged as likely purchasers, and more of those flags turn out to be wrong.

This report adopts the lower threshold on the assumption that the intended use case is identifying likely purchasers for follow-up (for example, targeted remarketing), where missing a genuine buyer is more costly than occasionally flagging a non-buyer. If the true cost of a false positive is high – for instance, if follow-up involves an expensive or intrusive action – a higher threshold, chosen with an explicit cost trade-off in mind, would be more appropriate. That decision ultimately depends on business context this dataset does not provide.

6 Conclusions

Which behaviours predict purchase. Exit rate is the dominant and most reliable behavioural predictor: each 1-percentage-point increase is associated with roughly a 25% reduction in the odds of purchase. Product-page engagement is a smaller, positive predictor (2.2% higher odds per 10 additional pages viewed), and `SpecialDay` carries a modest negative association.

Behaviour versus context. Behavioural variables – how a visitor engages with the site during their session – show larger, more consistent effects than contextual variables such as `Month`, `TrafficType`, or `VisitorType`. Several of these contextual effects are statistically significant, but with wider confidence intervals from smaller subgroup sizes; they read as descriptive patterns worth noting (e.g. the strong November conversion rate) rather than as robust, independent drivers of purchase.

Model performance. On held-out data, the final model achieves an AUC of 0.73, indicating moderate discriminating power. At a threshold chosen to balance sensitivity and specificity, it identifies about 78% of actual purchasers, at a precision of roughly 27% – a deliberate recall-over-precision trade-off, not a universal answer.

Limitations.

- `PageValues` was excluded as a leakage variable: it is calculated from pages that precede a transaction and would otherwise dominate the model without offering an actionable behavioural insight.
- 4.9% of sessions record product pages viewed in zero seconds; these were kept rather than dropped, on the judgement that this is a tracking artifact rather than invalid data.
- The outcome is heavily imbalanced (15.6% conversion rate), which is why accuracy is not reported as the headline metric and why the classification threshold was chosen deliberately rather than left at the default 0.5.
- This is observational data: the relationships reported here describe correlation with purchase, not proof of a causal driver of it.